

LECTURE 12

Figure 19.4: A Simplified Viral Replicative Cycle (Non-Enveloped DNA Virus)

This figure illustrates the reproduction process of a simple **DNA virus** that lacks an envelope and uses the host cell's machinery to replicate¹.

The Virus and Its Target

- The virus is an **intracellular parasite** that uses the equipment and small molecules of its host cell to replicate².
- In this simplest of viral cycles, the parasite is a **DNA virus** with a capsid consisting of a single type of protein³.
- The target is the **Host Cell**⁴.

The Replication Steps

1. **Entry and Uncoating:** The virus enters the host cell and is **uncoated**, releasing the **viral DNA** and **capsid proteins**⁵⁵⁵.
 2. **Genome Replication:** **Host enzymes** are used to **replicate the viral genome**, producing many new copies of the viral DNA⁶⁶⁶. The viral DNA serves as the template for the synthesis of new viral DNA⁷⁷⁷⁷.
 3. **Protein Synthesis:** Meanwhile, host enzymes **transcribe the viral genome into viral mRNA** (messenger RNA)⁸⁸⁸. **Host ribosomes** (the cell's protein-making machinery) then use this mRNA to **make more capsid proteins**⁹⁹⁹.
 4. **Assembly and Exit:** The newly synthesized **viral genomes and capsid proteins** **self-assemble** into new virus particles¹⁰¹⁰¹⁰. These newly formed viruses then **exit the cell** to find new host cells to infect¹¹¹¹¹¹.
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Figure: Infection by a Phage Triggers the CRISPR-Cas System (Bacterial Defense)

This figure illustrates how bacteria defend themselves against a **phage (bacteriophage)**, a virus that infects bacteria, using the **CRISPR-Cas** system. This system is a prokaryotic immune mechanism that utilizes a genetic record of past infections to specifically target and destroy the DNA of invading phages.

The CRISPR Region

- The **CRISPR region of DNA** in the bacterial cell contains segments of **DNA from previous infections** by the same type of phage, as well as from other phages, separated by **repeats**¹².

The Defense Steps

1. **Transcription: Infection by a phage** (the **invading phage**) triggers the **transcription** of the CRISPR region of the bacterial DNA into a long **RNA transcript**¹³.
2. **Processing:** The long RNA transcript is then **processed** into shorter **RNA strands**¹⁴. A key element is the **Complementary RNA** strand that is complementary to the DNA of the invading phage¹⁵.
3. **Complex Formation:** Each short RNA strand binds to a **Cas protein**, forming an active **complex**¹⁶. Multiple complexes are typically formed¹⁷.
4. **Targeting and Cutting:** The **Complementary RNA** within the complex acts as a guide, binding precisely to the matching **DNA from the invading phage**¹⁸. The **Cas protein** uses its **active sites that cut DNA** to then **cut the phage DNA**¹⁹.
5. **Degradation:** After being cut, the **entire phage DNA molecule is degraded** and can no longer be replicated by the bacterial machinery, effectively neutralizing the infection²⁰.

That's a great request! I can certainly provide detailed explanations of the images and content from the lecture slides you provided. The slides cover the basic structure, replication, and history of vaccination against viruses.

Slide 1: Introduction to Viruses - A Philosophical Question

This slide sets the stage by asking a profound question about the nature of life: **Are viruses "alive"?**¹

- The question is broken down by considering if an entity that has **DNA and RNA**² but **cannot make more copies of it** on its own³, and whose DNA/RNA codes for proteins⁴ but **cannot synthesize these encoded proteins**⁵, is truly "alive"⁶.
- This points to the core characteristic of viruses: they possess genetic material (DNA or RNA) and the instructions to make proteins, but they lack the necessary cellular machinery for replication and protein synthesis, making them obligate intracellular pathogens⁷.

Slide 2: The Structure and Diversity of Viruses

This slide illustrates the structural diversity of four different viruses and lists key facts about them.

- **Genetic Material and Protein Synthesis:** Viruses contain genetic information as either **DNA or RNA**. This genetic material is made into proteins by the machinery of the **host cell**⁸.
- **Host Range:** Viruses infect **ALL life forms**, including bacteria, plants, and animals⁹.
- **The "Chicken or the Egg" Question:** The slide poses an evolutionary question: Since viruses require a host cell to exist, which came first, the virus or the host cell?¹⁰
- **Examples of Virus Structures:**
 - **(a) Tobacco mosaic virus:** A rigid rod shape with a **helical capsid** and RNA inside, measuring about $18 \times 250 \text{ nm}$ ¹¹¹¹¹¹¹¹.
 - **(b) Adenoviruses:** Have an **icosahedral capsid** with a **glycoprotein spike** at each corner (vertex), measuring $70\text{-}90 \text{ nm}$ in diameter¹²¹²¹²¹².
 - **(c) Influenza viruses:** Possess eight different **RNA** molecules, each wrapped in a **helical capsid**. They have an outer **membranous envelope** studded with **glycoproteins**, and measure $80\text{-}200 \text{ nm}$ in diameter¹³¹³¹³¹³¹³¹³¹³¹³.
 - **(d) Bacteriophage T4:** A **complex capsid** with an **icosahedral head** (containing DNA) and a specialized **tail apparatus**, measuring about $80 \times 225 \text{ nm}$ ¹⁴¹⁴¹⁴¹⁴.

Slide 3: Simplified Viral Replicative Cycle (Non-enveloped)

This slide, along with its associated figure (Figure 19.4), details the replication of a simple **DNA virus without an envelope** (a non-enveloped virus) inside a host cell.

- **The Need for a Host:** To make more copies of itself, a virus particle (**virion**) must enter a cell where the machinery for genetic material replication and protein synthesis is functional¹⁵.
- **The Steps of Infection (Non-enveloped Virus):**
 1. **Entry and Uncoating:** The virus enters the host cell and is uncoated, releasing the **viral DNA** and **capsid proteins**¹⁶.
 2. **Genome Replication: Host enzymes** are "hijacked" and used to replicate the viral genome¹⁷¹⁷¹⁷¹⁷.
 3. **Protein Synthesis:** Meanwhile, host enzymes transcribe the viral genome into **viral mRNA**. **Host ribosomes** then use this mRNA to make more **capsid proteins**¹⁸.
 4. **Assembly and Exit:** The new **viral genomes** and **capsid proteins** **self-assemble** into new virus particles. These new virions then **rupture the host cell** and escape to find and infect other cells¹⁹¹⁹¹⁹¹⁹.
- **Species Specificity:** While viruses infect all life forms, each type of virus generally infects only a **specific species** or a specific **cell type** within that animal²⁰.
- **Species Barrier Crossing:** Excessive human intrusion into wildlife environments has been linked to an increase in the ability of viruses to **cross species barriers**²¹.

Slide 4: Viral Replicative Cycle (Enveloped RNA Virus)

This slide illustrates the replication of a more complex virus—one that uses **RNA as its genetic material** and has a **membranous envelope**.

- **Genetic Variety:** The slide notes that genetic information in viruses can be **DNA or RNA**, and can be **double- or single-stranded**²².
- **The Steps of Infection (Enveloped RNA Virus):**
 1. **Binding and Uptake:** **Glycoproteins** on the viral envelope bind to specific **receptor molecules** on the host cell, which promotes the viral uptake by the cell²³.
 2. **Uncoating:** The capsid and viral genome enter. **Cellular enzymes** digest the capsid, releasing the **viral genome (RNA)**²⁴.

3. **RNA Synthesis:** The viral genome RNA functions as a template for the synthesis of **complementary RNA strands** by a **viral RNA polymerase**²⁵. New copies of the viral genome RNA are then made using the complementary RNA strands as templates²⁶.
4. **Protein and Glycoprotein Synthesis:** The complementary RNA strands also function as **mRNA**. This is translated into:
 - **Capsid proteins** (in the cytosol)²⁷.
 - **Glycoproteins** for the viral envelope (in the **ER and Golgi apparatus**)²⁸.
5. **Assembly and Budding:** A capsid assembles around each viral genome molecule²⁹. Vesicles transport the envelope glycoproteins to the plasma membrane³⁰. The new virus then **buds from the cell**, acquiring its envelope, which is studded with viral glycoproteins, from the host cell's membrane³¹.

Slides 5-8: The History of Viral Vaccination

These slides detail the key historical milestones in the development of viral vaccines.

Dr. Edward Jenner and Smallpox (Slide 5)

- **Vaccination Concept:** Prior infection with a milder variant (like cowpox) provides **immunity** against subsequent exposure to a more virulent strain (smallpox)³².
- **Formal Demonstration:** Dr. Edward Jenner formally demonstrated this in the late 18th century, showing that inoculation with **cowpox** protected against **smallpox**³³. The term "vaccination" comes from *vacca*, the Latin word for cow³⁴.
- **Eradication:** The World Health Organization (WHO) declared smallpox eradicated in 1980³⁵.

Louis Pasteur and Rabies (Slide 6)

- **Attenuated Pathogens:** Louis Pasteur initially showed that **attenuated (weakened) pathogens** could protect farm animals against bacterial diseases like Cholera and Anthrax³⁶.

- **Rabies Work:** Around 1885, Pasteur used suspensions from rabies-infected, dried rabbit spinal cords to show that dogs could be protected from rabies³⁷.
- **Post-Exposure Prophylaxis:** His strategy was a **post-exposure vaccine prophylaxis**, a technique still used today³⁸. He famously "vaccinated" humans bitten by rabid dogs without proper animal trials because rabies was 100% fatal³⁹.
- **The Unseen Agent:** Pasteur developed his strategy without ever isolating the causative agent, which was later recognized as a **nonbacterial agent (a virus)**⁴⁰.

Max Theiler and Yellow Fever (Slides 7 & 8)

- **Yellow Fever:** This disease, caused by a **flavivirus** spread by the *Aedes aegypti* mosquito, was a global problem by the end of the 19th century⁴¹.
- **Viral Confirmation:** South African virologist **Max Theiler** showed that yellow fever was a **viral** disease⁴².
- **Attenuated Strain Development:** Theiler showed that repeated passage in mouse brain cultures reduced the virus's effect on most organs, though it risked increasing its impact on the central nervous system, which could cause encephalitis⁴³.
- **Vaccine 17D:** In **1937**, Theiler and Hugh Smith developed the **live attenuated yellow fever vaccine strain 17D**⁴⁴. This was achieved through 176 serial passages in various tissues, culminating in chick embryo tissue from which the brain and spinal cord had been removed⁴⁵.
- **Success:** The 17D strain lost its ability to cause nerve damage (**neurotropism**), organ damage (**viscerotropism**), and be transmitted by mosquitoes (**mosquito competence**), but still triggered an effective immune response⁴⁶. It was approved in **1938**, is safe and effective, and provides **lifelong protection with a single dose**⁴⁷⁴⁷⁴⁷⁴⁷.
- **Nobel Prize:** Theiler was awarded the **Nobel Prize in Physiology or Medicine in 1951** for his work on yellow fever, the first and only time the prize has been awarded for a vaccine⁴⁸.

Slides 9-11: Culturing Viruses

These slides focus on how viruses, which are **obligate intracellular pathogens** (meaning they cannot multiply on their own and need a live host cell and its machinery), are grown in a lab⁴⁹.

- [illegible]

Slides 12-13: Polio Virus and Vaccine Development

These slides trace the discovery of the polio virus and the development of the first successful polio vaccines.

- **Discovery of the Polio Virus:** In **1908**, Karl Landsteiner and Irwin Popper showed that polio was spread by a virus. They injected monkeys with a bacteriologically sterile suspension of spinal cord from a polio patient, which caused the monkeys to exhibit similar spinal cord lesions and develop paralysis⁵⁴⁵⁴⁵⁴⁵⁴.
- **Culturing Challenge:** Polio virus **could not be grown in chicken eggs**⁵⁵. Albert B. Sabin and Peter K. Olitsky grew it in human embryonic brain fragments, but this carried the risk of neurological damage if used for vaccine production⁵⁶.
- **Breakthrough: Thomas H. Weller, John F. Enders, and Frederick C. Robbins** (who won the 1954 Nobel Prize) made the critical discovery that poliovirus could grow in cultures of **various types of tissue** (not just neurons)⁵⁷⁵⁷⁵⁷⁵⁷. They successfully cultured the virus in skin, muscle, connective tissue, and intestinal tissue from human fetuses⁵⁸.
- **The Two Vaccines (c. 1950s):**
 - **Injected Vaccine:** Developed by **Jonas Salk**, containing **inactivated (killed) virus**⁵⁹.
 - **Oral Vaccine:** Developed by **Albert Sabin and colleagues**, containing **live attenuated (weakened) virus**⁶⁰.

Slide 14: The Concept of Immune Memory

This final slide summarizes the mechanism of vaccination and asks a question about the necessity of the whole virion.

- **Immune Response:** A whole virion that has "**forgotten**" its original virulence features is used to trigger an initial immune response⁶¹.
- **Immune Memory:** This initial response is stored in a dormant format as "**immune memory**"⁶².
- **Reactivation:** This stored, dormant memory **reactivates** when a new infection occurs with a virulent strain, providing protection⁶³.
- **The Final Question:** The slide concludes by asking if the whole virion is actually necessary to acquire immunity and protection against viral infections⁶⁴.